

EXHALE: A 1-D Multi-frequency Photoionization–Hydrodynamics Model of Escaping Exoplanet Atmospheres

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ABSTRACT

We present **EXHALE**, a one-dimensional, multi-frequency photoionization–hydrodynamics model for the XUV-driven escape of close-in exoplanet atmospheres and for the transit-absorption diagnostics that trace it. **EXHALE** evolves the compressible Euler equations for a single bulk fluid on a spherical grid with a Godunov finite-volume scheme (HLLC/Roe fluxes, PLM and WENO3 reconstruction) and relaxes to a stationary transonic wind, with an optional Jacobian-free Newton–Krylov (JFNK) solver that drives the discrete steady residual to machine-relevant levels. The thermal and ionization state is set by a coupled nonlinear system for hydrogen, helium, and the trace metals C, N, O, Mg, Ca, Na, and Fe, solved with a MINPACK dogleg iteration, or, for the systems for which an analytic Jacobian is written, a Newton iteration with that fallback: using Badnell radiative and dielectronic recombination, Voronov collisional ionization, and the Huang et al. charge-exchange compilation. Cooling includes Ly α , recombination, free–free, and closed-form metal-line coefficients fitted to CHIANTI. **EXHALE** solves the metastable He 2³S population that produces the He I 10830 Å triplet, builds the non-LTE $n = 2$ hydrogen population for the Balmer lines with an in-line Ly α escape-probability radiative-transfer field, allows diffusive He/H (and He/metal) separation, and can adopt a lower-atmosphere base from a photochemical (VULCAN) column. We apply the model to HD 209458 b, HD 189733 b, WASP-52 b, and WASP-121 b and compare synthetic He 10830, H α , Ly α , and metal-line transit depths against published observations. The four Newton-converged models span mass-loss rates $\log_{10}(\dot{M}/\text{g s}^{-1}) \approx 9.0\text{--}13.2$, from the compact HD 189733 b wind to the strongly irradiated WASP-121 b, with thermosphere temperature maxima of $\sim 0.8\text{--}1.2 \times 10^4$ K. The synthetic depths range from sub-percent H α on the quieter targets to tens of percent in He 10830 and Ly α , and are compared line by line with published measurements; the model reproduces the order of magnitude of the observed He 10830, H α , and Ly α absorption on the two cooler targets, over-predicts the WASP-52 b depths, and on WASP-121 b saturates against the truncation of the modeled domain, motivating the composition and domain refinements discussed in the text.

Keywords: planets and satellites: atmospheres — hydrodynamics — radiative transfer — methods: numerical

1. INTRODUCTION

Close-in giant planets receive intense stellar X-ray and extreme-ultraviolet (XUV) irradiation. Photoionization deposits energy in the upper atmosphere, heats it to $\sim 10^4$ K, and drives a transonic hydrodynamic wind that carries mass away from the planet (E. N. Parker 1958; R. A. Murray-Clay et al. 2009). This escape sculpts the mass–radius distribution of the exoplanet population and can be observed directly through excess absorption in a handful of spectral tracers during transit.

Four diagnostics dominate the observational picture. The He I 10830 Å triplet arises from the metastable He 2³S level, which is populated by recombination of He II and depopulated by photoionization, radiative decay, and collisions; because the mid-ultraviolet flux that photoionizes He 2³S and the XUV flux that produces He II depend on the host star, the 10830 line is a sensitive probe of both the wind structure and the stellar spectrum (A. Oklopčić & C. M. Hirata 2018; D. Yan et al. 2022). The hydrogen Balmer lines, chiefly H α , absorb from the $n = 2$ level, whose population is set

primarily by Ly α radiative pumping and thus requires a resonant-line radiative-transfer treatment (D. Christie et al. 2013; C. Huang et al. 2017). Ly α itself is a strong but interstellar-absorbed tracer of neutral hydrogen. Finally, metal resonance lines (Mg II h&k, Ca II H&K, Na I D, and Fe II) probe the ionization and thermal structure of the wind and, for the hottest planets, its metal content (C. Huang et al. 2023; T. T. Koskinen et al. 2022a).

A number of one-dimensional escape models have been built to interpret these tracers, spanning steady-state boundary-value solvers (R. A. Murray-Clay et al. 2009), time-relaxation photoionization-hydrodynamics codes (M. Salz et al. 2016; A. Caldiroli et al. 2021), grid-production schemes (D. Kubyshkina et al. 2018), and dedicated multispecies thermosphere models with excited-state microphysics (A. R. Taylor et al. 2025, 2026). EXHALE belongs to this last, diagnostic-oriented family. It began as a fork of the ATES photoionization hydrodynamics code (A. Caldiroli et al. 2021, 2022) and retains its Godunov hydro core and boundary/initial-condition skeleton, while adding a coupled trace-metal ionization network with self-consistent equation of state, CHIANTI metal-line cooling, the He 2³S metastable network, the non-LTE $n = 2$ hydrogen population with in-line Ly α radiative transfer, diffusive He/H separation, a direct steady-state (JFNK) solver, and a tiered coupling to the lower atmosphere. This paper documents the physics and numerics of EXHALE and applies it to four benchmark planets.

Section 2 describes the model. Section 3 contrasts EXHALE with ATES in both algorithm and physics. Section 4 presents the application to each planet (scaffold), and Section 5 summarizes.

2. METHOD

2.1. Hydrodynamics and numerical scheme

EXHALE solves the one-dimensional, spherically symmetric Euler equations for mass, radial momentum, and total energy of a single bulk fluid, with source terms for gravity, tidal (Roche) and centrifugal forces, photoionization heating, and radiative cooling. The discretization is a finite-volume Godunov method inherited from ATES: fluxes are computed with an approximate Riemann solver (HLLC in the runs presented here; Roe and local Lax–Friedrichs are also available, and the choice is a required input key with no compiled default), the interface states are reconstructed either with a piecewise-linear method (PLM, MC-limited) or with a third-order weighted ENO scheme using the weight functions of N. K. Yamaleev & M. H. Carpenter (2009) without their energy-stability term (input token WENO3), and time integration uses a strong-stability-preserving

Runge–Kutta update at CFL number 0.6 (A. Caldiroli et al. 2021; M. Salz et al. 2016). The grid is a mixed uniform–stretched mesh spanning the base radius to the outer boundary, with 500 cells plus ghost cells in the models presented here. Both the cell count and the spacing of the uniform region are runtime quantities; the latter matters because it sets the number of cells per base density scale height $H = k_B T / (\mu g)$, and a base resolved by too few cells supports a stationary cell-to-cell entropy (contact) mode that the non-dissipative Riemann fluxes do not damp. The code reports $H(T_{\text{eq}}) / \Delta r$ at startup; on the default grid this is ≈ 27 cells for HD 189733 b against ≈ 102 for WASP-121 b, and the high-gravity, compact HD 189733 b is correspondingly the planet of the four that needs a refined base. In this respect EXHALE and the PLUTO–CLOUDY interface of M. Salz et al. (2016) are numerical near-twins, sharing the same reconstruction, Riemann solver, and time integrator.

The stationary transonic wind is reached in two complementary ways. The default is time relaxation, converged on the spatial flatness of the mass flux $\rho v r^2$ over the wind window,

$$du = \frac{\max_j |\rho v r^2|_j - \min_j |\rho v r^2|_j}{\min_j |\rho v r^2|_j} < du_{\text{th}}, \quad (1)$$

with a two-stage schedule that marches first with PLM (robust) and then with WENO3 (low-diffusion) to a final tolerance $du_{\text{th}} = 10^{-3}$. This mass-flux criterion has two known blind spots: it cannot detect a uniform level drift of an already-flat profile, and on the strongly driven planets it dips transiently below threshold long before the wind settles, so a premature snapshot can be reported as steady. EXHALE therefore provides two further gates, both opt-in and disabled by default so that a run without them reproduces the legacy stop: (i) a level-stability gate on the windowed mean of $|\rho v r^2|$ and (ii) a diagnostic that evaluates the finite-volume steady residual $\mathbf{R}$ directly (mass, momentum, and heating-minus-cooling energy balance), which ranks candidate states independently of du and, when a tolerance is supplied, replaces du as the convergence test.

For quantitative mass-loss rates EXHALE provides a direct steady-state solver that finds the root of the discrete residual, $F(Y) = 0$, where $Y = (\rho, \rho v, E)$ on the physical cells and the ionization equilibrium (with its heating and cooling) is eliminated locally inside every residual evaluation. The nonlinear system is solved by pseudo-transient-continuation Newton iteration with an inexact Jacobian-free Newton–Krylov (JFNK) inner solve: a banded finite-difference Jacobian, obtained from a graph-colored stencil and factored with LAPACK, preconditions a matrix-free GMRES whose Jacobian–

vector products capture the weakly non-local radiation response. Robust convergence required a smooth ℓ_2 line-search merit, diagonal scaling across the several decades spanned by the base-to-wind density contrast, a differentiable base velocity valve, WENO smoothness weights frozen at the current iterate while the inner Newton model is built (the standard lagged-weights remedy, which keeps the strongly nonlinear weight response out of the Jacobian and the Krylov products) while the line search itself judges trial states on the residual with the weights recomputed, and a non-monotone (Grippe) step acceptance. The solver reaches steady residual levels unattainable by time marching at a small fraction of the cost; a best-iterate/fail-fast guard falls back to plain marching if the Newton basin is not reached, so an un-converged state cannot be mistaken for the answer.

The lower boundary is a fixed reservoir. Density and temperature are held at the base radius (total $n \sim 10^{14} \text{ cm}^{-3}$ near a μbar pressure level, T at the planet’s equilibrium value), and the ghost-cell velocity uses a one-way valve $v_{\text{gh}} = \max(v_1, 0)$ that admits wind-feeding inflow but suppresses the base oscillations (“breathing”) that otherwise develop at steep base density gradients: the same valve adopted independently by A. Caldiroli et al. (2021) and M. Salz et al. (2016). EXHALE adds a dynamic base-pressure correction that pins the base temperature exactly through the equation of state, and, for the Newton solver, a smooth variant of the valve. Pinning the ghost temperature to T_{eq} has no physical backing, however: the radiative equilibrium of the layer below is outside the model, so nothing in the code determines the ghost temperature, and the pin becomes a permanent contact discontinuity once the first interior cell settles far from T_{eq} . EXHALE therefore offers a second closure in which the ghost carries the first-cell temperature at the same pinned density, $p_{\text{gh}} = (n_{\text{tot}} + n_e)_{\text{bc}} T_1$, i.e. $dT/dr = 0$ at the base face; only the base density is then anchored. The T_{eq} pin remains the default, and the choice matters only for a planet whose base settles far from T_{eq} (Section 4.2). The upper boundary is free outflow, imposed as a zero-gradient copy of the last physical cell (linear extrapolation when the reconstruction is WENO3); no Mach condition is tested there, and whether the flow leaves supersonically is an outcome of the solution rather than a boundary condition. The outer radius defaults to the Hill/ L_1 radius of the Roche potential, and an explicit outer radius extends the domain past it while keeping the tidal potential (as for WASP-52 b, Section 4.3). Within physically realistic ranges the mass-loss rate is insensitive to the base density, temperature, and ionization fraction but scales steeply with the base radius,

$\dot{M} \propto R_p^3$ (R. A. Murray-Clay et al. 2009; M. Salz et al. 2016); EXHALE therefore treats the base radius (the pressure level it corresponds to) as the physically important lower-boundary quantity.

2.2. Photoionization, heating, and multi-frequency radiation

The stellar XUV field is attenuated radially and drives photoionization and photoelectric heating. EXHALE integrates each species’ energy-dependent photoionization cross section against the local stellar spectrum on a fixed energy grid. The spectrum may be read as a two-column (wavelength, flux) file (for example a MUSCLES spectral energy distribution scaled to the orbital distance (K. France et al. 2016)) or represented as a power law with a single spectral index but separate EUV and X-ray normalizations, each fixed by its band luminosity. Because the one-dimensional model is spherically symmetric and cannot resolve the day–night illumination geometry, the ionizing flux entering the hydrodynamics is reduced by a redistribution factor (the “Rate/2 + Mdot/2” approximation), valid as an average over the irradiated hemisphere. The same factor multiplies the stellar Ly α beam that pumps the $n = 2$ hydrogen population and the Balmer-continuum rates built on it (Section 2.5), so the excited-state diagnostics carry the same hemisphere average as the ground-state EUV rather than a second, inconsistent geometry.

The photoionization cross sections combine exact and fitted forms. Hydrogen and He II use the analytic hydrogenic cross section with the Gaunt-factor correction, exact at threshold ($6.3/Z^2$ Mb). The He I ground state uses the single-shell analytic fit of D. A. Verner et al. (1996, VF KY96) by default, which agrees with the legacy two-term ATEs fit to within 14% over 24.6–500 eV; the metals use the same VF KY96 functional form. The metastable He 2^3S level (the lower state of the 10830 triplet, whose threshold lies at ≈ 4.8 eV in the mid-ultraviolet) is not in the Verner tabulation, so it is represented by two VF KY96 terms (the $2s$ -shell channel and the resonance-averaged He II $n = 2$ channel) joined by a log-linear bridge across the Cooper minimum. This construction reproduces the D. W. Norcross (1971) tabulation shipped with **p-winds** (L. A. dos Santos et al. 2022) to within $\sim 10\%$ on the two fitted wings (degrading to 15% at the Cooper minimum itself, where the bridge joins them), and an independent TOPbase/Opacity Project R-matrix cross section to within $\sim 8\%$ in the smooth near-threshold region. The autoionizing resonances that TOPbase resolves above ~ 35 eV, and the smooth high-energy tail above the fitted cutoff, each change the He 2^3S photoionization rate by $\lesssim 1\%$

for the spectral slopes used here; because these are below the fit and composition uncertainties, they do not affect the 10830 prediction (a documented approximation valid for the smooth resonance-averaged background, which is what the broadband rate integral samples).

The energy deposited per photoionization is the photoelectron energy above threshold, summed over every absorber the ionization network carries: H I, He I, He II, the metastable $\text{He } 2^3S$, and the photoionizable trace metals (C, N, O, Mg, Ca, Na, and Fe ions); molecular runs add an H_2 photoionization term. The photoelectric heating efficiency is not unity: a fraction of each primary photoelectron’s energy drives secondary ionizations rather than heating the gas. For photoelectrons more than 40 eV above threshold **EXHALE** therefore applies the ionized-fraction partition of [J. M. Shull & M. E. van Steenberg \(1985\)](#) on both sides: the deposited heat is scaled by the heating fraction $f_{\text{heat}}(x)$, and the secondary H I and He I ionizations that the remaining energy pays for are added to the corresponding photoionization rates. The partition is on by default. Because from a cold initial condition its base feedback amplifies the startup transient into a runaway, it is activated in stages (the wind is first converged without it, the coupling is then switched on, and the run is converged again), so what is staged is when it is applied, not whether.

Beyond the direct photoelectric heating, **EXHALE** tracks four reprocessing channels that return trapped line and recombination energy to the gas. Balmer-continuum photoelectric heating draws on the excited $n = 2$ hydrogen population ([D. Christie et al. 2013](#); [C. Huang et al. 2017](#)). Ly α de-excitation heating returns the 10.2 eV excitation energy of Ly α -pumped $n = 2$ hydrogen as heat when the level is collisionally de-excited, in the trapped-Ly α picture of [C. Huang et al. \(2023\)](#); the pumping field is the in-line Ly α transfer of Section 2.5. Helium recombination-coupling heating captures helium recombination photons that photoionize and heat hydrogen. Finally, $\text{He } 2^3S + \text{H}$ Penning-ionization heating uses the temperature-dependent rate of [A. R. Taylor et al. \(2025\)](#), with a corresponding $\text{He } 2^3S + \text{H}_2$ term in molecular runs. The code writes each of these heating terms, together with the direct photoionization heating of every absorber, as a separate column of a diagnostic **Heating.breakdown.txt** file whose channel sum reproduces the total heating that enters the hydrodynamics (Table 1).

2.3. Coupled ionization and thermochemistry

The ionization state is computed in photoionization equilibrium as a single coupled nonlinear algebraic sys-

tem that includes hydrogen, helium, and the trace metals C, N, O, Mg, Ca, Na, and Fe, each carried through its relevant ionization stages, together with the metastable $\text{He } 2^3S$ level; the excited $n = 2$ hydrogen population is solved in a lagged-explicit outer iteration and enters the system as a frozen source. Solving all species simultaneously (rather than element by element) captures the shared free-electron density and the charge-exchange coupling between hydrogen and the metals. The system is solved with the MINPACK **hybrd1** dogleg method. For the systems for which an analytic Jacobian has been written (hydrogen alone, H/He, and H/He with metals) a Newton iteration with that Jacobian is used instead, falling back to **hybrd1** whenever it fails to converge; the metastable-triplet and molecular systems keep the MINPACK solve throughout, so the models presented here, which all carry $\text{He } 2^3S$, are solved by **hybrd1**.

The reaction rates are drawn from standard atomic-data sources, not fitted internally: radiative and dielectronic recombination from the Badnell data ([N. R. Badnell 2006](#); [N. R. Badnell et al. 2003](#)), collisional ionization from the [G. S. Voronov \(1997\)](#) fits, and charge exchange between hydrogen and the metals from the compilation of [C. Huang et al. \(2023, their Table 4\)](#), whose individual rate coefficients come from a range of sources, [J. B. Kingdon & G. J. Ferland \(1996\)](#) among them. Charge exchange between helium and hydrogen is applied in every system that carries helium, and the metal–helium and metal–metal reactions of the same table are available as an option. The trace-metal abundances are set at runtime through a configuration file, so a model can be run with or without metals without recompiling. Because the metals contribute to the mass, electron, and particle budgets, they are included self-consistently in the bulk-gas equation of state, so their presence feeds back on the base density, pressure, and mean molecular weight rather than being a passive overlay.

One entry of that tabulation is corrected here. As printed, the two oxygen rows of Table 4 of [C. Huang et al. \(2023\)](#) are inconsistent with detailed balance: the $\exp(-227/T)$ Boltzmann factor sits on the $\text{O}^+ + \text{H}$ row, whereas $\text{IP}(\text{O I}) = 13.6181$ eV exceeds $\text{IP}(\text{H I}) = 13.5984$ eV, so that $\text{O} + \text{H}^+ \rightarrow \text{O}^+ + \text{H}$ is the endothermic direction ($\Delta E/k = 227.7$ K, matching the printed argument to 0.3%) and is the one that must carry it. Read as printed the pair violates detailed balance by a factor of 1.47 at 8000 K; exchanging the two reactant labels restores it to within 9%. Both the expressions of [C. Huang et al. \(2023\)](#) and the two oxygen fits carried by **CLOUDY** ([C. M. Gunasekera et al. 2025](#)) descend from the same calculation ([P. C. Stancil et al. 1999](#)), and the

two sources attach the labels the opposite way, agreeing to 1–3% per direction under the exchange; this is therefore a check of the assignment and not an independent determination of the rate. Helium is the only other row of Table 4 carrying an explicit Boltzmann factor whose placement is fixed by the ionization potentials, and it is printed correctly, its argument 12.75 (in units of 10^4 K) matching $\text{IP}(\text{He I}) - \text{IP}(\text{H I}) = 12.752$. The most likely reading is therefore a transposition of the two oxygen labels rather than an error in the coefficients, and we adopt the exchanged assignment. This concerns the printed table only; we make no claim about the coefficients used in the calculations of [C. Huang et al. \(2023\)](#). Running our WASP-121 b model both ways leaves the wind unchanged for practical purposes (the mass-loss rate shifts by 0.004% and the temperature nowhere by more than 0.8 K), while the neutral-oxygen density in the ionized wind rises by 4% and O II falls by up to 18% near the base, so the assignment matters for an O I or O II line prediction rather than for the dynamics.

2.4. Cooling

The gas loses energy through several radiative channels, each of which EXHALE writes as a separate column of a diagnostic `Cooling_breakdown.txt` file whose channel sum reproduces the total radiative cooling. Collisional (line) excitation cooling is split by absorber into an H I term (dominated by $\text{Ly}\alpha$ and evaluated with the CHIANTI-consistent form of [R. Cen \(1992\)](#), consistent with [J. H. Black \(1981\)](#)), and separate He I and He II terms. Collisional-ionization cooling removes the ionization potential in each ionizing collision of H I, He I, and He II. Radiative recombination cooling of H II, He II, and He III uses case-B rates ([L. Hui & N. Y. Gnedin 1997](#)). Free-free (bremsstrahlung) emission scales as $T^{1/2} n_e n_{\text{ion}}$ times the free-free Gaunt factor of [P. A. M. van Hoof et al. \(2014\)](#), interpolated from their tabulation, and sums the contributions of hydrogen, helium, and the metal ions. The metastable helium level contributes a $2^3S \rightarrow 2^3P$ 10830 collisional-excitation cooling term. Metal-line emission is treated as described next.

Metal-line cooling uses closed-form analytic coefficients fitted to the CHIANTI database ([K. P. Dere et al. 1997](#); [G. Del Zanna et al. 2021](#)) rather than large run-time tables. A single-channel resonance line follows the exact two-level expression

$$\Lambda(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon(T) \Delta E e^{-\Delta E/k_B T}, \quad (2)$$

with the slowly varying effective collision strength $\Upsilon(T)$ fitted in the Burgess–Tully form; this reproduces the

Mg I, Mg II, Ca II, and Na I D cooling to a few percent. Species with many contributing transitions (Fe II and the C/N/O ions) are represented by a sum of effective two-level groups, $\Lambda(T) = T^{-1/2} \sum_i A_i e^{-T_i/T}$, whose group temperatures map onto physically distinct transition manifolds (infrared fine structure, optical metastable, ultraviolet resonance, and a high-energy tail). These CHIANTI fits replace an earlier set of C/N/O formulas inherited from AIOLOS that omitted nitrogen entirely and deviated for oxygen across the wind-launch temperature range; the fitted nitrogen cooling and corrected oxygen curves are genuinely new physics rather than a re-parameterization.

The CHIANTI fits carry the ground-term fine-structure lines in the optically thin, *low-density* coronal limit. Their critical densities are of order 10^0 – 10^5 cm^{-3} , decades below the base density of an irradiated atmosphere, so at the base the coronal form overstates what those lines can radiate by almost three to eight decades. For every C/N/O coolant whose ground term is split: C I $2p^2\ ^3P$, C II $2p^2\ ^2P$, N II $2p^2\ ^3P$, and O I $2p^4\ ^3P$, which is the complete set among the coolants: that term is therefore solved in exact statistical equilibrium at the local electron and neutral-hydrogen densities, and the CHIANTI curve is refitted to keep only the channels that leave the ground term. The solution saturates the lines at high density and collapses onto the coronal within-term value as the densities vanish, and because the collision rates carry both electron and hydrogen-atom terms it restores the hydrogen-collision channel that the electron-only coronal curves omit; its high-density limit reproduces the exact multilevel LTE emission. The hydrogen de-excitation rates are fits to published quantum scattering calculations: O I from [E. Abrahamsson et al. \(2007\)](#), C II from [Ġ. Barinova et al. \(2005\)](#), and C I and N II from [P.-G. Yan & J. F. Babb \(2023\)](#), in place of the approximate constants used previously.

All eight fine-structure lines with a transition probability are additionally allowed to be optically thick. Each line-center optical depth is accumulated over the column between the emitting cell and the top of the domain, and the corresponding escape probability $\beta(\tau)$: the plane-parallel Doppler form of [D. Hollenbach & C. F. McKee \(1979\)](#) and [T. de Jong et al. \(1980\)](#), renormalized by a factor two so that $\beta(0) = 1$ exactly, since radiation escaping downward is absorbed by the lower atmosphere and leaves the modeled gas either way: multiplies the spontaneous rate, $A_{ul} \rightarrow \beta_{ul} A_{ul}$, *inside* the statistical-equilibrium solution rather than as a factor on its result. Placing β inside matters: in the subcritical limit the cooling is fixed by the collisional excitation rate and must not depend on β , which an external factor

would violate, while in the saturated limit the escaping flux is proportional to β . In the tenuous wind $\beta \rightarrow 1$ and the optically thin result is recovered. Resonance metal lines that can be thick in a metal-rich wind (Mg II h&k) are still treated as optically thin; this is a documented limitation rather than a solved problem.

All of the CHIANTI coefficients are fitted over 10^3 – 10^5 K, and below that floor they are not merely unvalidated but unphysical: their softest exponentials carry excitation temperatures of several hundred to a few thousand kelvin that correspond to no ground-term splitting of the ion in question. The coronal part is therefore removed smoothly below 10^3 K by a cutoff factor that is exactly unity at and above the floor, with continuous value and temperature derivative (the bracketing temperature solve and the semi-implicit energy update both differentiate the cooling in T). Only the explicit statistical-equilibrium fine-structure terms, which are valid there, survive below the floor. The width of the cutoff is a modeling choice rather than a measurement, but it is no longer a consequential one: with the fine-structure lines saturated, the local radiative balance temperature of the base cell of all four models is unchanged to every digit over a factor 60 in that width, against a 500–650 K spread in the three models whose balance point lay below the fit floor when the coronal fine-structure floors were still in place (on WASP-121 b it lies above the floor, so the guard never engaged). In molecular runs an H_3^+ ro-vibrational cooling term is added (S. Miller et al. 2013).

The radiative channels above are the source term in the thermal energy equation; the gas also loses energy dynamically. Adiabatic (expansion) cooling (the $P dV$ work done by the expanding wind) is carried directly by the hydrodynamics rather than as a radiative loss, and it is typically the dominant energy sink in the outer wind, where it becomes comparable to or exceeds the radiative terms; the structure figures show it as a separate curve. Thermal conduction and viscous dissipation are implemented: the conductivity is the atomic-hydrogen form $\kappa(T) = 4.45 \times 10^4 (T/1000 \text{ K})^{0.7} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$ of A. J. Watson et al. (1981), with the viscosity tied to it by the monatomic Eucken relation, and both are evaluated by the same routine in the marching loop and in the steady residual, but both are off by default (the **Viscosity:** and **Conduction:** input keys), so neither enters the models presented here. Table 1 collects all of the heating, cooling, and dynamical channels together with their references.

2.5. The metastable helium triplet and in-line Ly α radiative transfer

The He I 10830 and Balmer diagnostics require the excited-state populations to be solved together with the wind. The metastable He 2^3S level is treated as a separate species in the coupled system. In steady state its abundance relative to He II is

$$\frac{n(\text{He } 2^3S)}{n_{\text{He}^+}} \simeq \frac{\alpha_3 n_e}{\Phi_3 + A_{31} + (q_{31a} + q_{31b})n_e + Q_{31}n_{\text{H}}}, \quad (3)$$

where α_3 is the recombination rate into 2^3S , Φ_3 the mid-ultraviolet photoionization rate, $A_{31} = 1.27 \times 10^{-4} \text{ s}^{-1}$ the radiative decay, and q_{31} , Q_{31} the electron and hydrogen collisional sinks. The hydrogen-collision sink includes the temperature-dependent Penning ionization rate of A. R. Taylor et al. (2025), which replaces the older constant value and is often the dominant 2^3S loss near the dense base.

The Balmer lines absorb from the $n = 2$ hydrogen population, which is built following the non-LTE rate equilibrium of D. Christie et al. (2013): collisional excitation, the recombination cascade (at a rate $\alpha_{2\ell} n_e n_{\text{H II}}$, with the case-B rate and its $2s/2p$ split taken from B. T. Draine 2011), two-photon decay, ℓ -mixing between $2s$ and $2p$, and (dominantly) Ly α radiative pumping $B_{1s \rightarrow 2p} \bar{J}_{\text{Ly}\alpha}$. Rather than a full Monte Carlo, which is intractable at the Ly α line-center optical depths of a hot Jupiter ($\tau_0 \sim 10^7$ – 10^8 , where core-skipping is both forbidden and incompatible with the core-dominated pumping), EXHALE evaluates $\bar{J}_{\text{Ly}\alpha}$ from a local escape-probability radiative-transfer field every time step. The treatment follows the Neufeld/Harrington wing-escape solution, which is accurate in the very optically thick limit $a\tau_0 \gg 1$ that applies here (D. A. Neufeld 1990). It combines an internal source term (recombination cascade plus electron-impact excitation) closed on the $2p$ balance and a $(1 - \beta)$ factor that prevents the source function from diverging as the gas ionizes in the outer wind, plus a trapped stellar-beam term whose penetration through the broad stellar-line wings uses the stellar line width. Two quantities enter that stellar term and take the same values in all four models presented here. The stellar Ly α half-width is $\Delta v_\star = 70 \text{ km s}^{-1}$ (input key **Ly α stellar halfwidth**), the broad plateau of C. Huang et al. (2017), and it sets how deep the beam reaches through the wings: the penetrating fraction is $\mathcal{T}_\star = \text{erfc}[x_1 v_{\text{th}}/(\sqrt{2} \Delta v_\star)]$ at the wing-escape frequency x_1 , with v_{th} the local thermal speed. The scattered beam then builds up above free streaming by a bounded factor $1 + (b - 1)(1 - \beta)\mathcal{T}_\star$ with $b = 5$ (**Ly α stellar boost**), which tends to b where the beam is fully trapped and to unity both in the thin outer wind and at the pen-

Table 1. Heating, cooling, and dynamical energy channels in EXHALE. Each radiative channel is written to a separate column of `Heating_breakdown.txt` or `Cooling_breakdown.txt`, whose channel sums reproduce the total heating and cooling that enter the hydrodynamics.

Channel	Reference / form
Heating	
Photoionization, H I	photoelectron energy above threshold
Photoionization, He I, He II	
Photoionization, He 2^3S metastable	mid-UV, $\gtrsim 4.8$ eV
Photoionization, C/N/O/Mg/Ca/Na/Fe ions	
Photoionization, H ₂	molecular runs
Lyman–Werner photodissociation, H ₂	B. T. Draine & F. Bertoldi (1996); molecular runs
Balmer-continuum photoelectric ($n = 2$ H)	D. Christie et al. (2013); C. Huang et al. (2017)
Ly α de-excitation heating	C. Huang et al. (2023); Sec. 2.5
He recombination-coupling heating of H	
He 2^3S +H Penning ionization	A. R. Taylor et al. (2025)
He 2^3S +H ₂ Penning ionization	molecular runs
Radiative cooling	
Collisional excitation, H I (Ly α)	R. Cen (1992); J. H. Black (1981)
Collisional excitation, He I, He II	
Collisional ionization, H I/He I/He II	
Radiative recombination (case B)	L. Hui & N. Y. Gnedin (1997)
Free-free, H + He + metal ions	$T^{1/2} \times$ Gaunt factor, P. A. M. van Hoof et al. (2014)
He 2^3S metastable $2^3S \rightarrow 2^3P$ 10830	
Metal lines, C/N/O/Mg/Ca/Na/Fe	K. P. Dere et al. (1997); G. Del Zanna et al. (2021)
H ₃ ⁺ ro-vibrational	S. Miller et al. (2013); molecular runs
Dynamical (energy equation)	
Adiabatic (expansion) cooling	$P dV$ work of the wind
Thermal conduction, viscous dissipation	A. J. Watson et al. (1981); off by default

etration edge, where the unbounded $1/\beta$ buildup would diverge. The value $b = 5$ is calibrated against Figure 11 of C. Huang et al. (2023) rather than derived, and is the one tuned quantity in the Ly α field. The $2p$ balance is closed against $A_{2p \rightarrow 1s}\beta + D$, with $\beta = \pi^{-1/4}\sqrt{a/\tau}$ the wing escape probability and D collecting the channels that destroy a trapped Ly α photon without re-emitting it: collisional de-excitation, photoionization of $n = 2$, and ℓ -mixing followed by two-photon decay. Only the first of these returns the 10.2 eV to the electron gas; that collisional de-excitation heating of the radiatively pumped $n = 2$ population is included by default, as the correction to the one-way coronal excitation rate that supplies the H I line cooling. No escape-probability

factor is applied to that cooling itself, which appears to be the right form here: measured on these models, of order 98% or more of the trapped Ly α photons escape, and the collisional destruction that could suppress the collisionally driven cooling reaches at most $\sim 0.06\%$. A Sobolev (wind-velocity) escape channel is combined with the static wing escape unconditionally. It changes $\bar{J}_{\text{Ly}\alpha}$ by less than a percent *where the $n = 2$ population resides*: weighted by n_{2p} the mean change is below 0.4% on all four models, and 90% of the $n_{2p}r^2$ column of HD 189733 b sits inside $1.11 R_p$, because there the velocity gradient is small and the line is optically thick. Only in the thin outer wind, beyond $\sim 2\text{--}4 R_p$, does the Sobolev term overtake the static escape and dominate

the closure, and there the $n = 2$ population it would pump is smaller by orders of magnitude.

Two routes to an externally computed field are available. The escape model itself can read a tabulated radial profile $\bar{J}_{\text{Ly}\alpha}(r)$ in place of the in-line closure (input key `Jlya RT file`), which leaves the wind solution spherically symmetric. For percent-level $\text{H}\alpha$ work under one-sided illumination, where the pumping field is only cylindrically symmetric, a separate post-processor instead reads a two-dimensional scattering-rate grid $P_\alpha(\varpi, z)$ from the `LaRT` Monte Carlo code (K.-i. Seon & C.-G. Kim 2020), rebuilds $n_{2p}(\varpi, z)$ from it, and integrates the $\text{H}\alpha$ transit along the line of sight; the wind structure it reads is still the spherical one, and the $\text{He} 10830$ transit, set by the spherically symmetric $\text{He} 2^3S$ balance, is unaffected.

2.6. Diffusive separation

Because helium is four times heavier than hydrogen, it is under-dragged by the escaping hydrogen wind, so the escaping He/H ratio can fall below the bulk abundance with increasing altitude. `EXHALE` retains its single bulk-momentum formulation but represents this separation with a molecular- plus eddy-diffusion term that lets the He/H ratio vary with radius, in the spirit of A. R. Taylor et al. (2025) and consistent with the multi-fluid result of L. Xing et al. (2023). The binary molecular diffusion coefficient is the He-in-H expression of P. M. Banks & G. Kockarts (1973), $D = 1.52 \times 10^{18} (1/m_{\text{H}} + 1/m_{\text{He}})^{1/2} T^{1/2} / n_{\text{tot}} \text{ cm}^2 \text{ s}^{-1}$, and a metal in hydrogen uses the same form with its own mass. The homopause where the diffusive separation sets in is controlled by an eddy-diffusion coefficient that can be supplied from the lower-atmosphere model (Section 2.7). The same framework extends to He/metal separation. This altitude-dependent composition is what the transit post-processor requires for a physically consistent $\text{He} 10830$ column; it is optional and off by default.

2.7. Lower-atmosphere boundary

The base of the escape model sits near a μbar pressure level, below which the atmosphere is denser, cooler, and possibly molecular. `EXHALE` couples to this region through an optional handoff file (`base.inp`) that overrides the base temperature, base radius, He/H ratio, and eddy-diffusion coefficient before the derived constants are computed. A tiered set of lower-atmosphere models writes this file: an analytic hydrostatic column that returns the $1 \mu\text{bar}$ radius and a molecular-hydrogen fraction; a molecular-chemistry option with H_3^+ cooling for genuinely cool sub-Neptunes; and, at the most complete tier, a photochemical column from the

`VULCAN` code (S.-M. Tsai et al. 2017), whose disequilibrium composition replaces the chemical-equilibrium estimate (C. Visscher et al. 2006; J. I. Moses et al. 2011). For a warm hot Jupiter the photochemical base dissociates far more hydrogen than chemical equilibrium predicts, so the atomic base that the escape model assumes rests on photochemistry rather than on thermal dissociation; `VULCAN`, however, does not release the atomic metals (Na/Mg/Ca/Fe), whose abundances remain user-supplied.

The coupling is one-way by default (lower atmosphere $\rightarrow$ base $\rightarrow$ escape). Two caveats bound its validity. First, the composition must not be double-counted: if the handoff file sets the He/H ratio from the lower model, the metal abundances should be that model’s atomic-release values rather than solar totals. Second, and most important, the handoff assumes that the base pressure lands below the thermospheric temperature rise. If it instead falls inside a thermospheric temperature inversion (where the escape model’s own heating has already raised the temperature) then adopting a cool lower-atmosphere T and composition at that level double-counts the transition region and misrepresents the base state. The handoff is therefore valid only when the chosen pressure level sits beneath the onset of thermospheric heating; for dayside-hot or strongly irradiated cases the base radius from an isothermal or radiative-equilibrium column is a lower bound that must be checked against the escape model’s own temperature structure.

3. DIFFERENCES FROM ATES

`EXHALE` inherits its hydrodynamic core and boundary/initial-condition skeleton from `ATES` (A. Caldiroli et al. 2021, 2022): the Godunov finite-volume scheme, the SSPRK time integrator, the choice of HLLC, Roe, and LLF fluxes, PLM and WENO3 reconstruction, the 500-cell mixed grid, the two-stage PLM $\rightarrow$ WENO3 relaxation to $\Delta\dot{M}/\dot{M} < 10^{-3}$, the fixed-reservoir lower boundary with the one-way velocity valve, the zero-gradient outflow upper boundary, and the cold neutral isothermal initial condition with a linear velocity seed. The ionization is likewise solved in photoionization equilibrium with a MINPACK method, and ion advection is applied as a post-processing step. What `EXHALE` adds is in the microphysics and in the steady-state solver, not in the hydrodynamic backbone. Table 2 summarizes the differences, which are then discussed.

Solver.—`ATES` reaches steady state by time relaxation only, so its convergence is the mass-flux flatness du ; `EXHALE` keeps that path but adds a direct steady-state

Table 2. Algorithm and physics: EXHALE compared with ATES v2.0

Axis	ATES v2.0	EXHALE
Hydro core	Godunov, HLLC/Roe/LLF, PLM/WENO3, SSPRK	Same (inherited)
Steady state	Time relaxation only (<i>du</i> criterion)	Time relaxation + JFNK Newton finish; residual-based stop
Energy update	Explicit source	Semi-implicit energy + Brent <i>T</i> -solve
Metals	None	C, N, O, Mg, Ca, Na, Fe in the coupled system, in the EOS
Ionization solve	MINPACK dogleg	Analytic-Jacobian Newton + MINPACK fallback
Recombination	Standard tables	Badnell RR + DR
Cooling	Ly α , recomb, free-free (H/He)	Adds CHIANTI metal-line fits (with fine-structure saturation)
He triplet	He 2^3S in the equilibrium network	Same, extended: coupled to the metals and to the diffusive He/H split; temperature-dependent Penning rate
Balmer / Ly α RT	Not solved	$n = 2$ non-LTE + in-line Ly α escape-probability field
Charge exchange	-	Huang et al. (2023) Table 4: H $\leftrightarrow$ metal and He $\leftrightarrow$ H by default, metal $\leftrightarrow$ He and metal $\leftrightarrow$ metal optional
Diffusion	None (fixed He/H)	Optional He/H and He/metal diffusive separation
Lower atmosphere	Fixed base	Tiered base (analytic / molecular / VULCAN handoff)
Transit spectra	He 10830, Ly α (TPM.py)	Adds H α , H β , metal doublets; optional Roche-triaxial geometry

solver (Section 2.1) that drives the discrete residual toward zero, a stricter standard that also exposed cases where the *du* criterion stops at a premature, non-steady snapshot. EXHALE also replaces the explicit source update with a semi-implicit energy step and a bracketing (Brent) temperature solve that avoids the hot-root trap of stiff metal cooling.

Metal chemistry.—The central physics addition is the coupled trace-metal network. ATES carries hydrogen and helium only; EXHALE solves C, N, O, Mg, Ca, Na, and Fe through their ionization stages in the same nonlinear system, with Badnell radiative and dielectronic recombination, Voronov collisional ionization, and the C. Huang et al. (2023) charge-exchange network, and it feeds the metals back into the equation of state. This enables both the metal-line cooling and the metal transit diagnostics that ATES cannot produce.

Excited states and radiative transfer.—ATES already solves the He 2^3S metastable level inside its equilibrium network and synthesizes the He 10830 and Ly α transits

from it. EXHALE keeps that level but couples it to the metals and to the diffusive He/H split, and adds the non-LTE $n = 2$ hydrogen population for the Balmer lines together with the in-line Ly α escape-probability field that pumps it: neither of which is in ATES. These, with the diffusive He/H separation and the tiered lower-atmosphere coupling, turn the inherited escape solver into a line-diagnostic model. Notably, the base oscillation (“breathing”) phenomenology and its mitigations (the smooth valve, the advection guard) are EXHALE-era findings not present in the ATES description.

4. RESULTS

We apply EXHALE to four transiting planets that span the relevant parameter range and have published transit-absorption measurements. For each we report the Newton-converged wind structure (Figures 1–4), the steady-state mass-loss rate, and the synthetic transit depths against observations (Figures 5–7). All four solutions are converged with the JFNK steady-state solver rather than stopped on the *du* criterion, reaching discrete steady residuals of 1.8×10^{-4} , 8.3×10^{-4} , 8.7×10^{-4}

and 7.8×10^{-4} for HD 209458 b, HD 189733 b, WASP-52 b and WASP-121 b; this matters quantitatively, because a du -threshold stop alone carries a path-dependent spread of several percent in $\dot{M}$.

Quoted transit depths are the line-center excess absorption of the model curve, with the instrument-convolved value given after a slash where the convolution changes it appreciably. The line-spread convolution uses the resolving power of the instrument each line is compared against: $R = 8 \times 10^4$ for CARMENES (HD 189733 b He 10830), 9×10^4 for CRIRES+ (WASP-121 b), 2.5×10^4 for Keck/NIRSPEC (WASP-52 b), 6×10^4 for HET/HRS (H α), 1.4×10^5 for ESPRESSO (WASP-52 b H α), and 1.5×10^4 for HST/STIS G140M (Ly α), and the metal doublets, which have no counterpart observation here, keep the code defaults. Using one fixed resolving power instead changes the convolved depths by less than 1.5% except for the two lowest-resolution comparisons, where it inflates them by 6–8%. The figures additionally include the broadening by the (tidally locked) planet rotation, which lowers the plotted peak further; the planet, stellar, and spin parameters all come from each run directory’s own input file. Of the four models only HD 189733 b carries the diffusive He/H and He/metal separation of Section 2.6; the other three are run with a fixed He/H ratio, so composition gradients are not a differentiating factor between them. The structure figures resolve both the radiative cooling and the heating into their individual channels, with the collisional-excitation cooling split by absorber into an H I (Ly α -dominated) term and the helium terms.

4.1. HD 209458 b

EXHALE relaxes HD 209458 b to a steady transonic wind with a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.47$. The temperature climbs from the ~ 1450 K base through a steep thermosphere to a broad maximum of $\approx 7.7 \times 10^3$ K near 1.4 – $1.6 R_p$, and the gas becomes predominantly ionized above $\approx 1.3 R_p$, where H II overtakes H I; helium follows, singly ionized stages exceeding the neutral above $\approx 1.4 R_p$ (Figure 1). The outflow is slow and subsonic across the modeled thermosphere, reaching 0.8 km s^{-1} at the $2 R_p$ escape radius and crossing the sonic point only at $\approx 4.1 R_p$, well outside it. In the cooling budget the Ly α -dominated collisional excitation and recombination channels balance the photoionization heating in the ionized layer, while adiabatic expansion becomes comparable to the radiative losses in the outer wind.

The synthetic He 10830 triplet has a line-center depth of 10.2/9.6% and Ly α reaches 25.1/23.1%, while H α is weak at 0.31% (Figures 5, 6, and 7). The metal reso-

nance lines of this solar-composition run are predicted at 11.8/6.9% (Mg II h&k), 3.6% (Ca II H&K), and 1.8/1.6% (Na I D); the large convolution effect on Mg II reflects how narrow its saturated core is. The Ly α comparison is the one clear failure of this model. The core is inaccessible to observation because of interstellar and geocoronal absorption, so the measurement of A. Vidal-Madjar et al. (2003) constrains the wings, and there the model falls an order of magnitude short: at $|v| = 50$ – 120 km s^{-1} the digitized profile absorbs 3–15% while the model absorbs 0.7–1.7%. Integrated over the 1215.15–1216.10 Å domain (-130 to $+100 \text{ km s}^{-1}$ in the stellar rest frame) on which A. Vidal-Madjar et al. (2003) measure a $15 \pm 4\%$ decrease, the model gives 5.0%, low by 2.5σ . The model Ly α line is far too narrow: a one-dimensional spherical wind produces no population of fast neutral hydrogen, which is what the observed wing absorption requires and what radiation-pressure acceleration and stellar-wind charge exchange are invoked to supply. For H α , A. G. Jensen et al. (2012) report a mid-transit absorption of $M_{\text{abs}} = (0.50 \pm 0.80) \times 10^{-4}$ over their 16 Å band; the model gives 0.85×10^{-4} on the same band, within 0.4σ of that measurement. The digitized trace itself shows a broad antisymmetric structure, significant in the original analysis, that the narrow model line does not reproduce.

4.2. HD 189733 b

For the denser, more strongly bound HD 189733 b the model gives a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.14$, the lowest of the four. The thermosphere is hotter and more compact than that of HD 209458 b, with a temperature maximum of $\approx 1.23 \times 10^4$ K, and the wind is again slow and subsonic through the modeled region, reaching only 0.5 km s^{-1} at $2 R_p$ (Figure 2). Hydrogen and helium ionize over a narrow shell just above the base; the diffusive He/H separation included in this run lets the He/H ratio fall with altitude.

This is the planet whose lower boundary needs both of the base treatments of Section 2.1. On the default spacing its base density scale height spans only ~ 27 cells, the least margin of the four, and the resulting stationary cell-to-cell entropy mode survives in the converged state, so the model quoted here doubles the base resolution. Refinement alone does not remove the alternating signature of the first cells, because that part of it measures the ghost-to-cell jump imposed by pinning the ghost temperature to T_{eq} rather than the entropy mode; this run therefore also uses the continuous-temperature ghost closure ($dT/dr = 0$ at the base face). The two together leave a converged base whose ghost and first cell agree to 0.05% in temperature and whose density

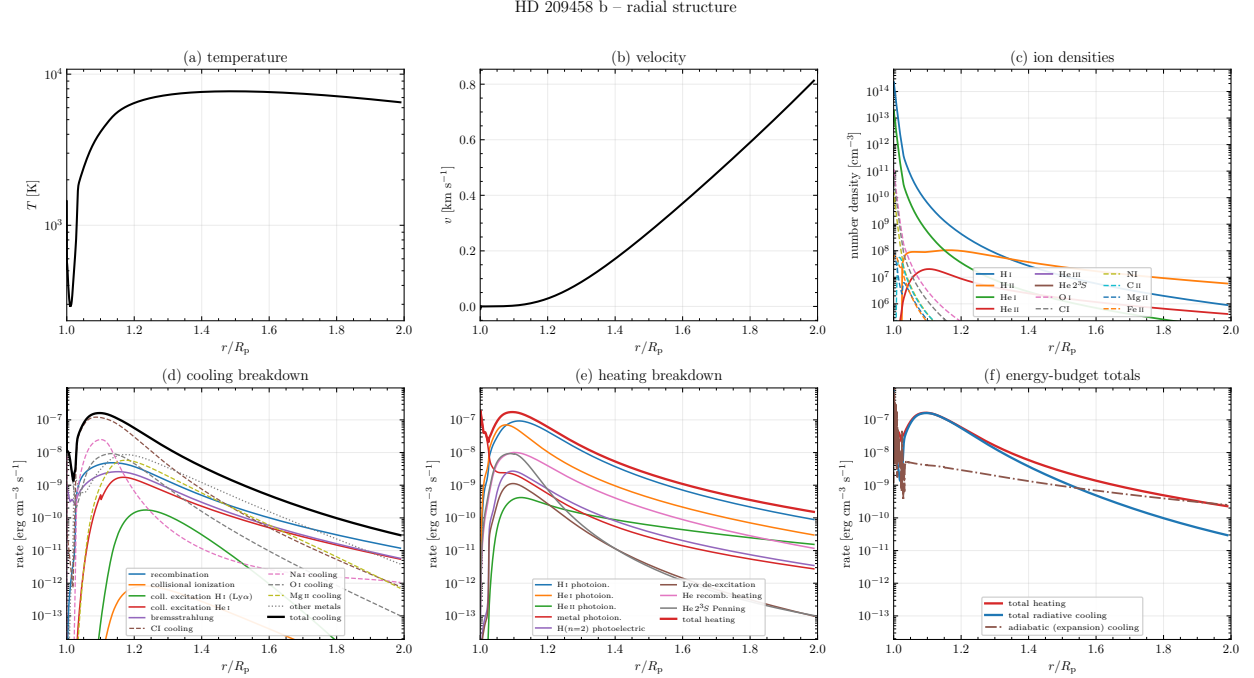


Figure 1. Radial structure of the converged HD 209458 b model: (a) temperature, (b) radial velocity, (c) ion number densities (H and He stages, the metastable $\text{He } 2^3S$, and the most abundant metal ions in the wind), (d) the radiative cooling breakdown, (e) the heating breakdown, and (f) the energy-budget totals. Panel (d) shows each radiative cooling channel from the model output, with the collisional-excitation cooling split by absorber into H I (the $\text{Ly}\alpha$ -dominated H-line cooling), He I, and He II. Panel (e) shows the photoionization heating of each absorber (H I, He I, He II, $\text{He } 2^3S$, and the metals), the excited-hydrogen photoelectric and $\text{Ly}\alpha$ de-excitation heating, and the He-recombination and $\text{He } 2^3S$ Penning heating. Panel (f) compares the total heating, the total radiative cooling, and the adiabatic (expansion) cooling computed from the velocity field. The individual channels in panels (d)–(f) are the ones tabulated in Table 1. Curves are drawn to the $2 R_p$ escape radius. Generated by `python/make_struct_figures.py`.

inversion is gone ($\rho_1/\rho_{\text{gh}} = 0.95$, against 2.33 under the T_{eq} pin). What survives is a residual velocity alternation of order 10^2 cm s^{-1} over the first few cells, decaying to a few cm s^{-1} by the tenth, which appears to be a numerical artifact of the lower boundary rather than a physical feature. The base temperature this run settles to, $\sim 560 \text{ K}$, is not a local radiative balance: once the ground-term fine-structure lines are saturated (Section 2.4) the total radiative cooling of the first cell falls some two orders of magnitude below the local heating, so the temperature there is set by the inner boundary and by the flow rather than by the cooling function. The model contains no stellar optical/infrared absorption and no thermal background, so nothing enforces a floor at the equilibrium temperature for a layer that has shielded out the XUV.

The synthetic $\text{He } 10830$ depth is 2.9/2.8% and $\text{H}\alpha$ is 0.61%, while $\text{Ly}\alpha$ reaches 37.9/36.7% (Figures 5, 6, and 7). The comparison with M. Salz et al. (2018) depends on which measure is used, so we quote the integrated one: the model equivalent width is $16.0 \text{ m}\text{\AA}$ against their $12.7 \pm 0.6 \text{ m}\text{\AA}$, i.e. 1.26 times larger, while

the line-center depths differ by more (2.9% against their peak $1.04 \pm 0.09\%$; their band-averaged $0.88 \pm 0.04\%$ over $\pm 10 \text{ km s}^{-1}$ is the number usually quoted). The model does not reproduce the observed triplet ratio: it puts 4.45 times more absorption in the 10830 component than in 10829, against the measured 2.8 ± 0.2 , which M. Salz et al. (2018) read as the signature of an optically thick, compact absorber. For $\text{H}\alpha$, A. G. Jensen et al. (2012) measure $M_{\text{abs}} = (-8.7 \pm 1.5) \times 10^{-4}$ over $16 \text{ \AA}$ ($W_\lambda = 0.0141 \text{ \AA}$ after their stellar-variability correction), while the model gives 2.5×10^{-4} ($W_\lambda = 0.0041 \text{ \AA}$), 3.5 times shallower. So the metastable helium column is reproduced to within about a quarter while the Balmer absorption is short by a factor of a few: the opposite sign of the two mismatches points to the excited-state microphysics rather than to the overall wind density.

4.3. WASP-52 b

WASP-52 b is a low-gravity, strongly irradiated planet, and the model produces a correspondingly large mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 11.70$. The temperature reaches $\approx 9.3 \times 10^3 \text{ K}$ and the gas is highly ionized

HD 189733 b – radial structure

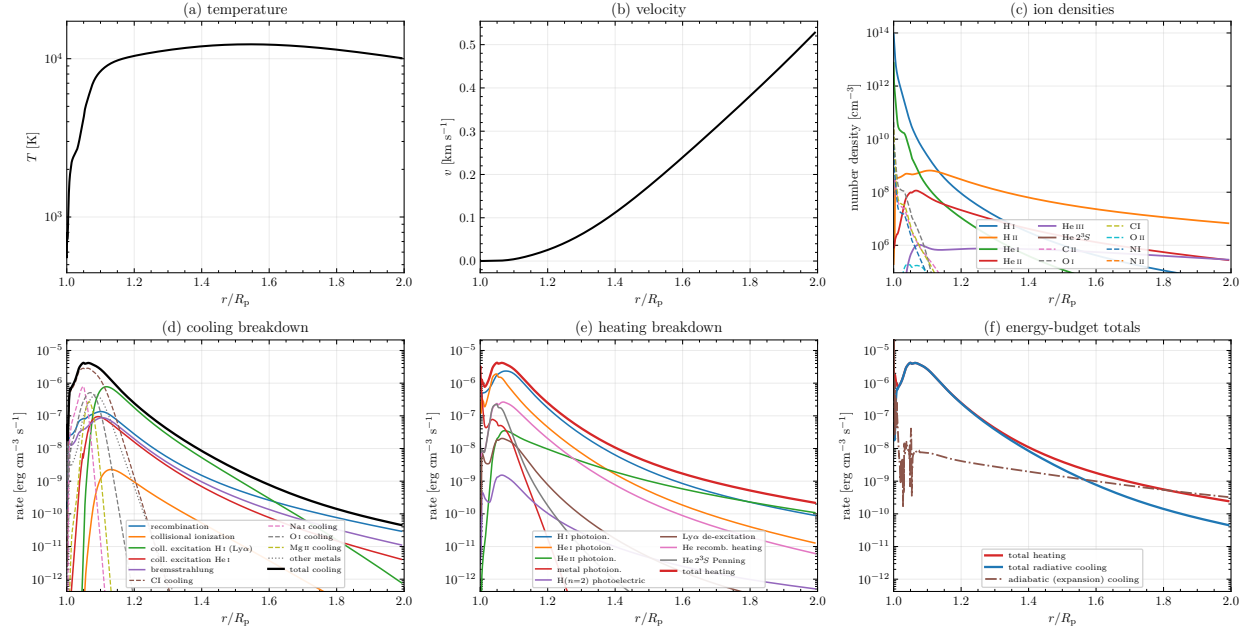


Figure 2. Radial structure of the converged HD 189733 b model, as in Figure 1. This run includes the diffusive He/H and He/metal separation. Generated by `python/make_struct_figures.py`.

through the extended thermosphere, which is modeled here on a domain reaching to $10 R_p$; the wind accelerates smoothly, passes 10 km s^{-1} at the $2 R_p$ escape radius, and crosses the sonic point just outside it at $\approx 2.2 R_p$ (Figure 3). The low assumed He/H ratio (≈ 0.02) still leaves a large neutral-hydrogen and metastable-helium column along the extended line of sight.

The resulting transit depths are the largest of the sample: the model He 10830 triplet reaches 25.2/23.2%, H α 4.4%, and Ly α is fully saturated near 100% (Figures 5–7). The metal lines are deeper still, with Mg II at 71.2/62.8% and Ca II at 16.9%. These greatly exceed the measured absorption. The J. Kirk et al. (2022) He 10830 spectrum peaks near 3.6%, and G. Chen et al. (2020) report a firm H α detection with a line contrast of $0.86 \pm 0.13\%$ (a measured depth, not an upper limit) against which the modeled 4.4% is about five times too deep. The most likely contributors are the large radial extent of the absorbing column and the assumed composition. The hydrodynamic domain here runs to $10 R_p$, and the transit integration carries each line of sight out to $R_*/R_p = 6.19 R_p$, where the stellar disk ends; both are far outside the L_1 radius of $2.53 R_p$. A reduced He/H ratio and a line of sight truncated nearer the Roche lobe would both lower the predicted depths, and quantifying this sensitivity is left to future work rather than tuned here. A companion analysis of the same target separates the two levers. Holding the microphysics and

composition fixed and truncating the line-of-sight integration at the L_1 Roche point, rather than carrying it to the stellar-disk edge, lowers the synthetic He 10830 depth by a factor of 3–10 on its own, because most of the absorbing column on the extended domain lies beyond the radius the planet can gravitationally retain. Reducing the helium abundance from solar to H/He $\approx 98/2$ then accounts for a further factor of about two, and the two together reproduce the measured depth. The larger of the two effects is therefore the domain, which argues that the over-prediction reported here is primarily a geometry problem, to be addressed with a Roche-lobe truncation of the transit integration rather than by adjusting the microphysics.

4.4. WASP-121 b

WASP-121 b is the most extreme target, an ultra-hot Jupiter close to Roche-lobe overflow. The model, computed without diffusive separation, gives a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 13.20$, the highest of the sample. The thermosphere is very hot (of order 10^4 K through most of the modeled shell and rising steeply toward the $1.5 R_p$ escape radius), and the wind reaches 14 km s^{-1} there, crossing the sonic point at $\approx 1.6 R_p$, just beyond the escape radius (Figure 4). The gas is strongly ionized, with metal ions such as C II, O II, and N II carrying an appreciable fraction of the electrons and contributing to the metal-line cooling. Because the escape radius lies inside the Roche geometry, the profile near the

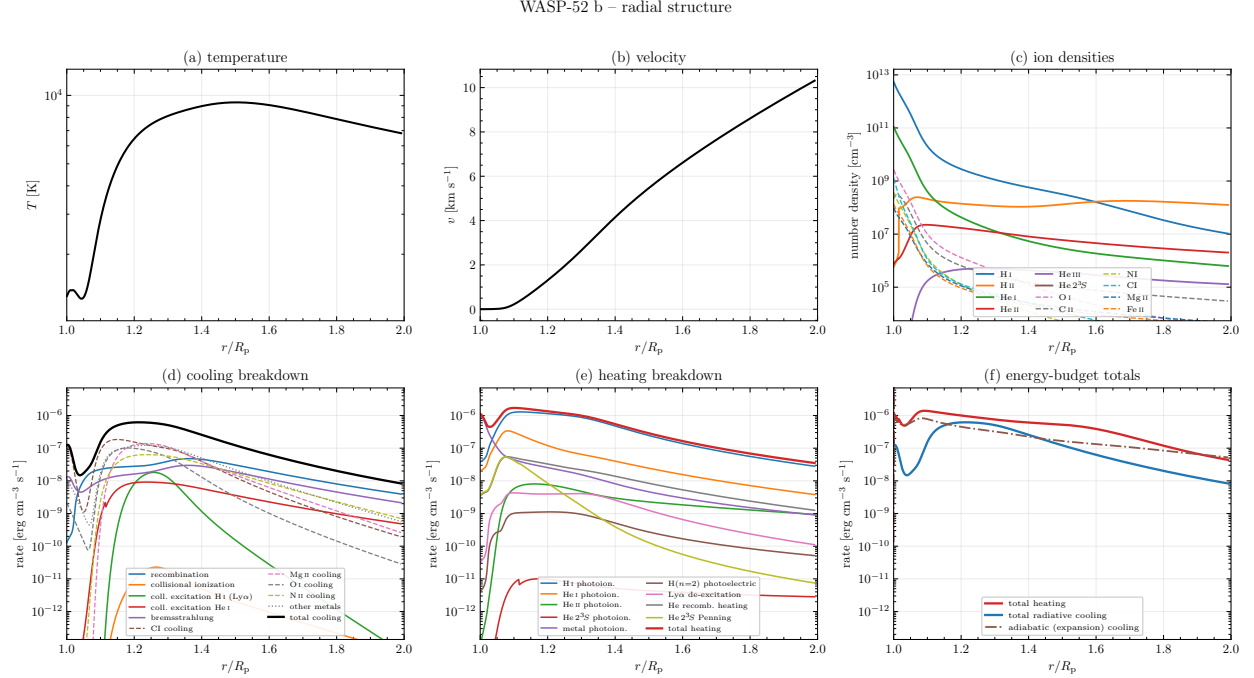


Figure 3. Radial structure of the converged WASP-52 b model, as in Figure 1, shown to the $2 R_p$ escape radius (the computational domain extends to $10 R_p$). Generated by `python/make_struct_figures.py`.

outer boundary is sensitive to the domain truncation, and the solution shows a boundary-layer temperature feature there.

The synthetic depths are the most nearly uniform across lines of the four planets, and several of them sit at the same value: He 10830 is 3.7%, H α 3.4%, Ly α 3.8%, Mg II 3.8/3.7%, and Ca II 3.7% (Figures 5–7). That common ceiling is geometric rather than physical. A line whose absorbing column is optically thick all the way to the edge of the modeled profile blocks the whole annulus between the planet and that edge, so its depth cannot exceed $(R_p/R_\star)^2 (r_{\text{out}}^2 - 1)$, which is 3.9% here for a profile ending at $r_{\text{out}} = 1.61 R_p$ on a $1.458 R_\odot$ star. The saturated lines sit just under that bound. Every one of these depths is therefore a lower bound imposed by the domain rather than a prediction, and none of them should be read as a measurement of how much absorption the wind actually produces: what the model establishes on this planet is that the wind is optically thick in all five lines out to the truncation radius. A quantitative depth requires a domain carried past the Roche lobe, which is where the one-dimensional spherical geometry stops being the right description in any case. For the same reason, the fact that the He 10830 depth exceeds the S. Czesla et al. (2024) measurement, which peaks near 2%, is not yet evidence of a discrepancy in the microphysics. We note only two features of the comparison. The observed core sits slightly red-

ward of the model line: its absorption-weighted centroid over $\pm 20 \text{ km s}^{-1}$ of the 10830 component is $+1.5 \text{ km s}^{-1}$ against -1.2 km s^{-1} for the model, the “slight redshift” S. Czesla et al. (2024) report and read as the signature of an asymmetric, tidally shaped outflow. And the modeled mid-transit equivalent width, 62.6 mÅ , is about twice the transit-averaged $\approx 30 \text{ mÅ}$ they quote, although a mid-transit value and a transit average are not the same measure and part of that ratio is expected. That the metal lines saturate at all is consistent with the metal-rich, highly ionized wind expected for this planet (C. Huang et al. 2023; T. T. Koskinen et al. 2022a).

5. SUMMARY

We have described EXHALE, a one-dimensional multi-frequency photoionization–hydrodynamics model for escaping exoplanet atmospheres. Building on the ATES Godunov hydro core, EXHALE adds a coupled H/He/metal ionization network with Badnell recombination, Voronov collisional ionization, and the C. Huang et al. (2023) charge-exchange compilation; CHIANTI metal-line cooling; the He 2^3S metastable triplet; the non-LTE $n = 2$ hydrogen population with an in-line Ly α escape-probability field; optional diffusive He/H separation; a direct JFNK steady-state solver; and a tiered lower-atmosphere coupling. Applied to HD 209458 b, HD 189733 b, WASP-52 b, and WASP-121 b, the model produces mass-loss rates and synthetic He 10830, H α , Ly α , and metal-line transit spectra for direct compari-

WASP-121 b – radial structure

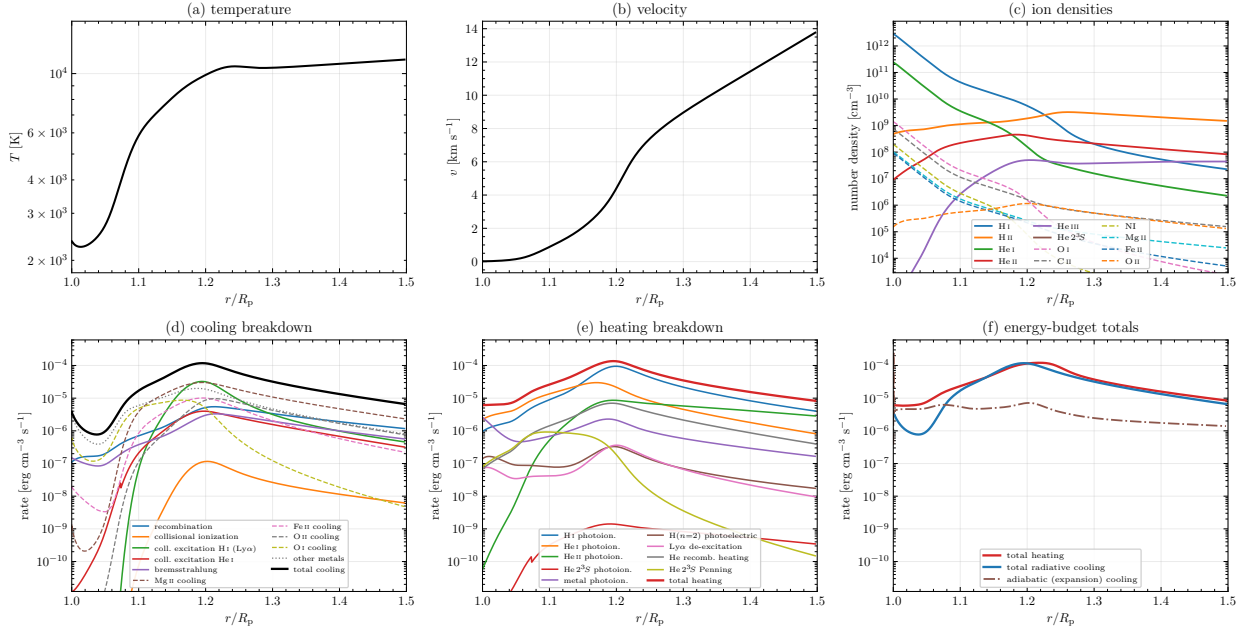


Figure 4. Radial structure of the WASP-121 b model (no diffusive separation), as in Figure 1, shown to the $1.5 R_p$ escape radius. The temperature panel is clipped to a robust upper percentile so the boundary-layer feature near the escape radius does not dominate the scale. Generated by `python/make_struct_figures.py`.

son with observations. Across the four benchmarks the model gives mass-loss rates of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.47$ (HD 209458 b), 9.14 (HD 189733 b), 11.70 (WASP-52 b), and 13.20 (WASP-121 b), each converged with the steady-state solver rather than stopped on the mass-flux criterion, and synthetic line depths that match the order of magnitude of the measured He 10830, $\text{H}\alpha$, and $\text{Ly}\alpha$ absorption on HD 209458 b and HD 189733 b, while over-predicting the WASP-52 b depths and saturating against the domain truncation on WASP-121 b. The priorities these point to are the composition and the line-of-sight domain, a transit geometry carried past the Roche lobe for the most extended winds, and the excited-state microphysics behind the opposite-sign HD 189733 b He 10830 and $\text{H}\alpha$ mismatches.

Transit transmission: He I 10830 Å

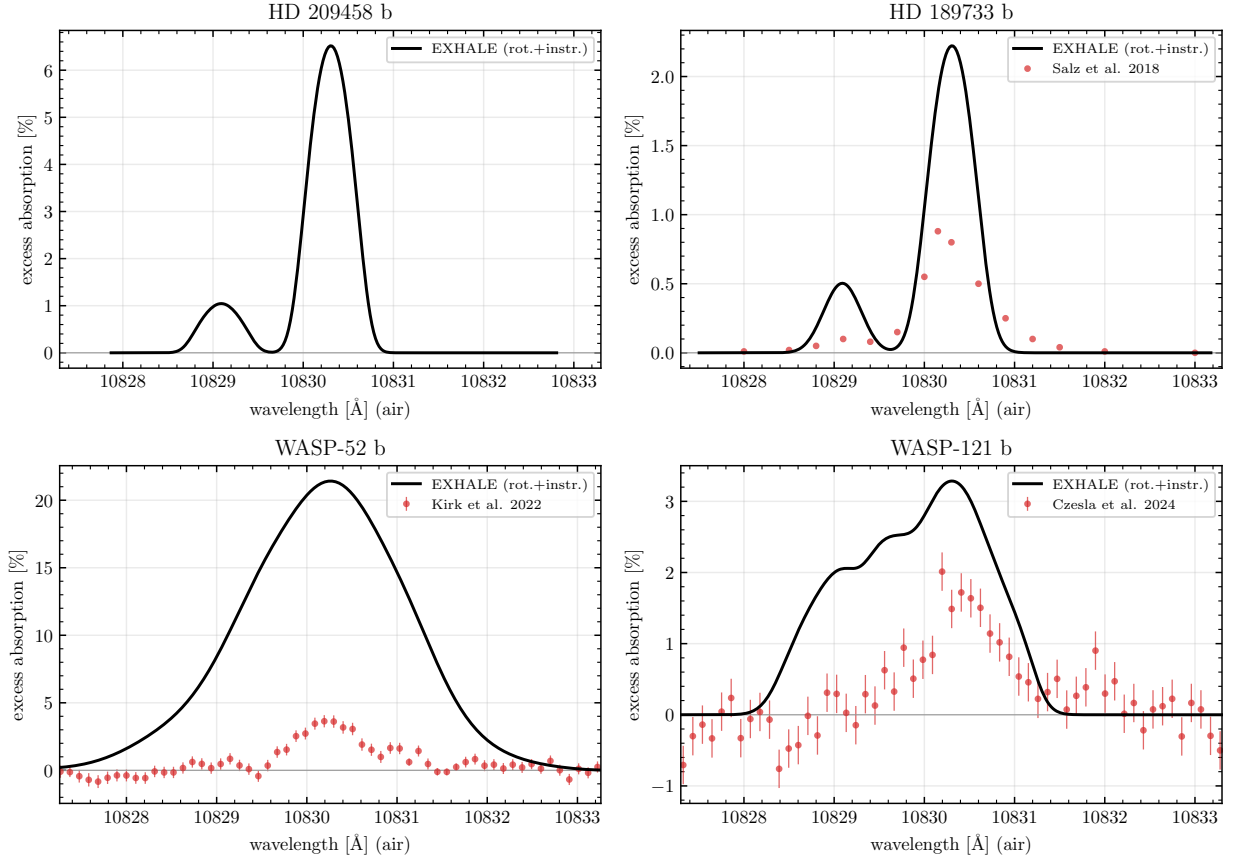


Figure 5. Synthetic He I 10830 transit spectra (rotation- and instrument-convolved) for the four planets, compared with published observations where available: M. Salz et al. (2018) for HD 189733 b, J. Kirk et al. (2022) for WASP-52 b, and S. Czesla et al. (2024) for WASP-121 b. All curves are shown as excess absorption, $(1 - T)$ in percent; each observation is converted to this common axis from its native convention (dF/F, excess-absorption percent, or normalized in/out flux). Generated by `python/make_transit_figures.py`.

Transit transmission: $H\alpha$ 6562.8 Å

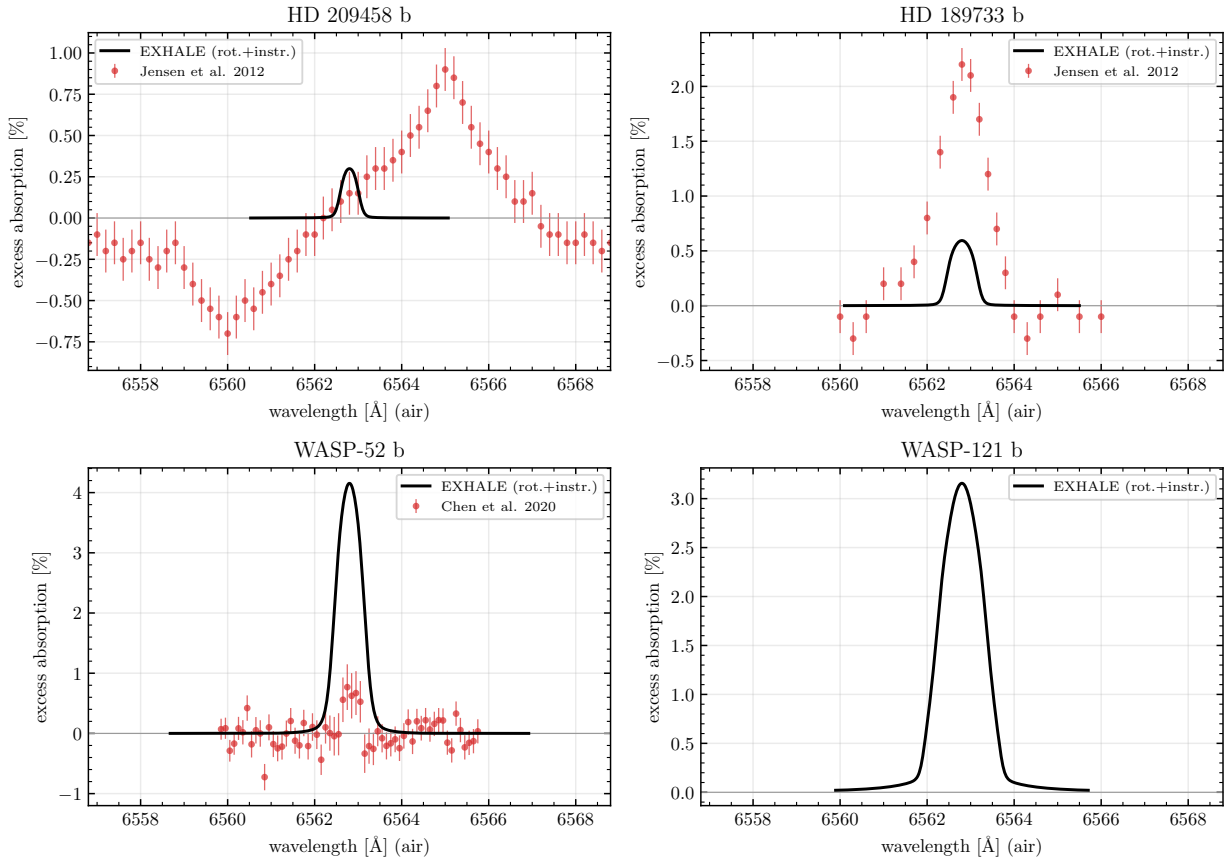


Figure 6. Synthetic $H\alpha$ transit spectra, as in Figure 5, compared with the digitized A. G. Jensen et al. (2012) traces (HD 209458 b and HD 189733 b) and the G. Chen et al. (2020) ESPRESSO spectrum (WASP-52 b). Generated by `python/make_transit_figures.py`.

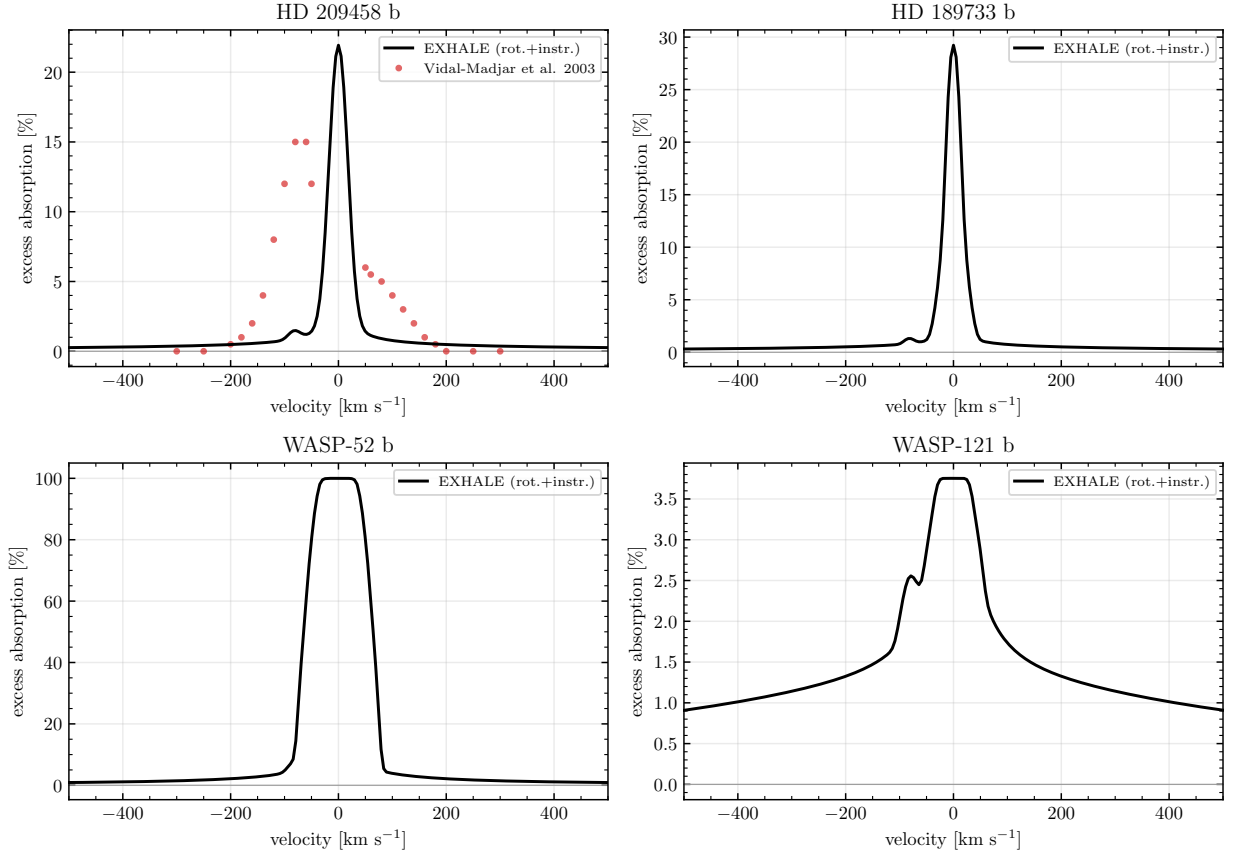
Transit transmission: $\text{Ly}\alpha$ 1215.67 Å

Figure 7. Synthetic $\text{Ly}\alpha$ transit spectra on a velocity axis, as in Figure 5. The HD 209458 b panel overplots the wing profile of A. Vidal-Madjar et al. (2003); the line core (roughly $\pm 40 \text{ km s}^{-1}$) is omitted in the data because of interstellar and geocoronal absorption. No $\text{Ly}\alpha$ measurements are available for the other three planets. Generated by `python/make_transit_figures.py`.

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